⛅ 6TH GRADE SCIENCE: WEATHER & CLIMATE REVIEW

🌦️ Weather vs. Climate S6E4

	WEATHER	CLIMATE
Time frame	Right now, today, this week	Averaged over **30+ years**
Changes	Quickly — hour to hour	Slowly, over a long time
Example	"It's storming this afternoon."	"Summers here are hot and humid."

🧠 **Memory trick:** Weather is your **mood** (changes in an afternoon); climate is your **personality** (your usual pattern). One rainy day doesn't change Georgia's warm, humid climate!

🌐 Layers of the Atmosphere S6E4

The atmosphere = the blanket of gases around Earth: mostly **nitrogen (78%)** and **oxygen (21%)**.

Layer (low → high)	What's There
Troposphere	The lowest layer; **we live here and ALL weather happens here** ⛈️
Stratosphere	Holds the **ozone layer**; jets fly here for a smoother ride ✈️
Mesosphere	Coldest layer; most **meteors burn up** here ☄️
Thermosphere	Very hot; **auroras** and the space station 🚀
Exosphere	The outermost layer; fades into outer space

🧴 Ozone Layer

In the stratosphere; absorbs most of the Sun's harmful **UV rays** — sunscreen for the whole planet.

🌡️ Greenhouse Gases

CO_2, methane, water vapor — they **trap heat** like a blanket: sunlight in, some heat kept from escaping. Without them Earth would freeze; too much and Earth warms.

✏️ ACTIVITY 1: Layer It On!

Which layer? (troposphere, stratosphere, mesosphere, thermosphere, exosphere)

1. All weather happens here: ________________
2. Contains the ozone layer: ________________
3. Meteors burn up here: ________________
4. Auroras & the space station: ________________
5. Fades into outer space: ________________
6. The layer where YOU live: ________________

☀️ Uneven Heating → Wind S6E4a,b

All weather starts here: **the Sun does NOT heat Earth evenly.**

LAND 🏖️

Heats up **fast**, cools down **fast** (hot sand at noon, cold sand at night)

WATER 🌊

Heats up **slowly**, cools down **slowly** (the ocean stays cool at noon, warm at night)

The equator gets **direct** sunlight (hot); the poles get slanted sunlight (cold). **How wind forms:** warm air gets lighter and **RISES** → cooler, heavier air flows in to take its place → that moving air IS the wind!

💨 **Air pressure** = the weight of air pressing down. Warm rising air = **LOW** pressure; cool sinking air = **HIGH** pressure. Wind always flows from **HIGH** → **LOW**. A **sea breeze**: day-heated land makes warm air rise; a cool breeze flows in from the water.

✏️ ACTIVITY 2: Go With the Flow

1. Warm air rises / sinks (circle one)
2. Wind flows from ______ pressure to ______ pressure
3. Heats faster in the sun: land / water (circle one)
4. Warm rising air creates ______ pressure
5. The equator is hot because sunlight there is ____________
6. A cool daytime wind from sea to shore: ________________

⛅ *Flip over for fronts, storms & hurricanes! →*

⛅ WEATHER & CLIMATE REVIEW (continued)

🌀 Air Masses, Fronts & Storms S6E4d

Air mass = a large body of air with similar temperature & moisture. Where two different air masses meet = a **FRONT** — and fronts are where most exciting weather happens!

Front	What Happens	Typical Weather
Cold Front	Cold air pushes **UNDER** warm air, lifting it **quickly**	**Sudden, strong storms**; then cooler, clearer air
Warm Front	Warm air slides slowly **OVER** cold air	**Long, light, steady rain**; then warmer air

⛈️ Thunderstorms & Tornadoes

Warm, **moist air rising fast** (often along a cold front) builds tall storm clouds → heavy rain, lightning, thunder. If the rising air starts to **spin** → possible **tornado**: a violently rotating column of air touching the ground. Georgia gets both, especially in spring!

📉 Pressure → Weather

LOW pressure = stormy (warm rising air forms clouds & rain).
HIGH pressure = clear (cool sinking air clears the skies).

✏️ ACTIVITY 3: Cold Front or Warm Front?

Write **C** (cold front) or **W** (warm front):

____ 1. Cold air shoves under warm air
____ 2. Brings long, light, steady rain
____ 3. Brings sudden, strong storms
____ 4. Warm air slides up and over cold air
____ 5. Moves fast; cooler clear air follows
____ 6. Moves slowly; warmer air follows

🌀 Hurricanes — Powered by Warm Ocean Water S6E4

A **hurricane** = Earth's largest, most powerful storm: a giant spinning system of thunderstorms. Its energy source: **warm ocean water**. The Sun heats the ocean → warm water **evaporates** → the warm, moist air rises and cools, releasing huge energy that feeds the storm. The calm center is the **eye**.

⛽ **The ocean is the fuel tank:** hurricanes form over warm tropical oceans, strengthen over water, and **weaken over land** — they lose their warm-water fuel! (Connects to Ch. 3: evaporation is the fuel line.)

✏️ ACTIVITY 4: Storm Chaser Quiz

1. A hurricane's energy comes from: ________________
2. The calm center of a hurricane: ________________
3. Hurricanes weaken over land because: ________________
4. A violently rotating column of air: ________________
5. Low pressure usually means ____________ weather
6. Storms with lightning need warm, ________ air rising fast

🔗 The Big Chain of Weather

Sun heats Earth UNEVENLY → warm air RISES (low pressure) → cool air flows in = WIND → air masses meet at FRONTS → clouds & STORMS → warm oceans fuel HURRICANES

⭐ Almost every weather question on the test traces back to **uneven heating by the Sun**. Start there!

📝 I Can… (Self-Check)

☐ Tell weather (now) from climate (30+ yr average)
☐ Name the 5 atmosphere layers and what's in each
☐ Explain what the ozone layer and greenhouse gases do
☐ Explain why land and water heat unevenly
☐ Explain how wind forms (warm rises, cool flows in)
☐ Connect pressure to weather (low = stormy, high = clear)
☐ Compare cold fronts and warm fronts
☐ Explain where hurricanes get their energy

🎯 Based on Georgia Standards of Excellence: S6E4 — You're ready for Milestones! 💪